

Abundances in atmospheres of Ap-stars: HD 188041 (V1291 Aql), HD 111133 (EP Vir), HD 118022 (78 Vir), HD 204411 and HD 110066 (AX CVn)

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Introduction

HD 188041, HD 111133, HD 118022, and HD 110066 are magnetic chemical peculiar stars of Ap (p - peculiar) spectral class. Significant exceed of Cr, Sr and Eu are main elements to determining magnetic Ap stars. These stars belong to a compact group of stars with similar strengths of surface magnetic field 3–4 kGs. HD 204411 is an evolved Ap star with small magnetic field which does not exceed 1 kG. Spectra of Ap stars are characterized by anomalous chemical abundances of many elements. Usually light elements, He, CNO, show a deficiency compared to the solar abundances while heavier elements, especially rare earth elements (lanthanoids), are enhanced by few orders of magnitudes [2].

To explain the anomalous chemical compositions in magnetic stars, in 1970 Michaud [1] proposed a mechanism of diffusion separation in the atmospheres of Ap stars under the combined action of several forces, the main of which are gravitational pressure directed toward the center of the star, and the radiation pressure force that pushes particles into the outer layers of the atmosphere.

Abundance calculations

The element abundances were determined using the lines of different ionization stages: neutral/first ions (C to Ba) and first/second ions for lanthanides (La to Yb). The abundances were evaluated using the equivalent widths of the corresponding spectral lines measured in high resolution, high signal-to-noise spectra. The parameters of stellar model atmospheres, determined in the frame of the present work, are given in Table 1. Abundance calculations were performed with the WidSyn code [4] which takes the magnetic field into account. Line atomic parameters were extracted from the Vienna Atomic Line Database VALD3 [3].

Results

Stellar abundances patterns are displayed in Fig. 1. Average abundances derived from the lines of consecutive ionization stages (neutral and first ions or first and second ions for REE) are shown by filled and open circles, respectively. For HD 188041, HD 111133, HD 118022 and HD 110066 element compositions are similar within the errors of the determination, and the abundance patterns correspond to the general observed anomalies in Ap-stars: CNO deficiency, practically solar Na and Mg abundance, 1-2 dex overabundance of the iron peak elements, and a large excess of the rare-earth elements (REEs).

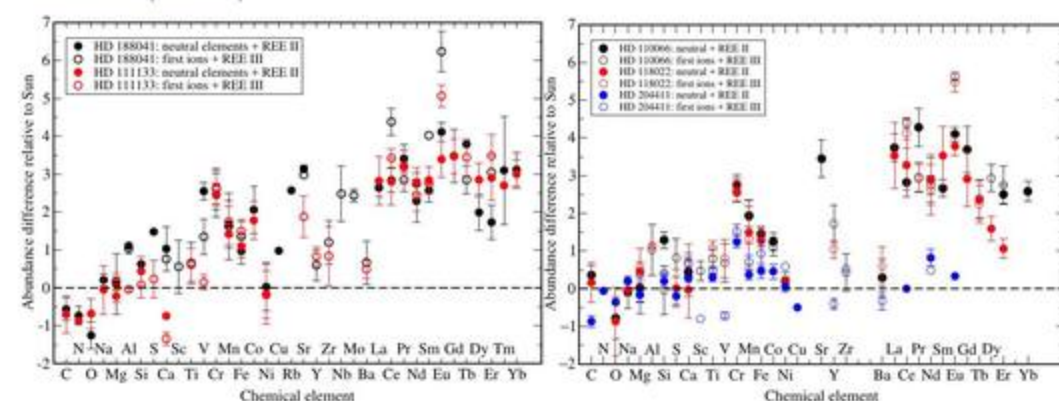
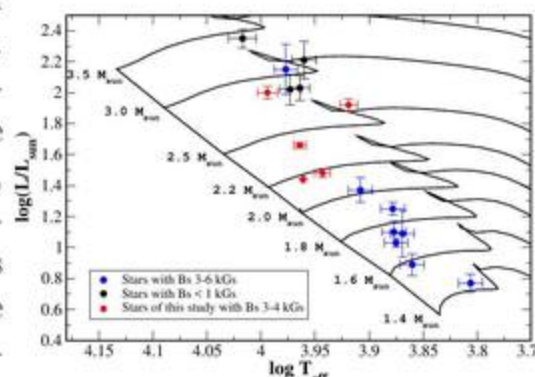


Figure 1: Abundances of HD 188041, HD 111133 (left panel), and HD 110066, HD 118022, HD 204411 (right panel) relative to the solar photospheric values.

These abundances were used in the determination of the fundamental parameters.

Table 1 lists the fundamental parameters of Ap stars derived from the detailed self-consistent spectroscopic analysis taking into account abundance anomalies.

Fig. 2 shows the positions of program stars (red circles) on the HR diagram. Four stars that have similar abundance patterns are lying in the same area (middle part) in HR diagram, while HD 204411 that shows the deficiency of REE compared with 4 stars and also has the smallest value of the magnetic field in the sample, is located in the end of Main Sequence strip on HR diagram. Position of Figure 2: Stars positions on the Hertzsprung-Russell diagram. Stars from previous studies are indicated by black (for magnetic field < 1 kG) and blue (for magnetic field 3–6 kG) circles.



Stars	T_{eff} , K	$\log g$	B_s , kG	$\log L/L_\odot$
HD 111133	9770(200)	4.0(2)	4.0	2.00(4)
HD 188041	8770(150)	4.2(1)	3.6	1.48(3)
HD 118022	9142(30)	4.2(2)	3.0	1.44(1)
HD 110066	9200(100)	4.1(1)	4.1	1.66(2)
HD 204411	8300(150)	3.6(1)	≤ 0.8	1.92(4)

Table 1: Fundamental parameters of Ap stars. Error of the measurement in the last digits is given in parentheses.

Two peaks in distribution of elements are clearly visible in the abundance distribution of Ap stars (Fig. 1). First peak is associated with iron group elements, and the second one with the rare earths. The first peak is observed in almost all Ap stars, and the second peak is observed in the stars which have not yet evolved. Interestingly that Ba which precedes REE in periodic Mendeleev table is closed to solar or even slightly deficient in almost all Ap stars.

Conclusions

A comparative analysis of the average chemical abundance in the atmospheres of magnetic Ap stars is presented. Atmospheric abundances of these stars are similar within the errors of the determination, and the abundance patterns correspond to the generally observed anomalies in Ap-stars. There is deficiency of light elements of carbon, nitrogen and oxygen (CNO), practically solar Na and Mg abundance, 1-2 dex overabundance of the iron peak elements, and a large excess of the lanthanides (rare-earth elements). In evolved star HD 204411, the iron peak elements show smaller anomalies, and the lanthanide abundances decrease up to solar values.

References

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